end part of the hydraulic brake system is called as slave cylinder. It is made up of cast iron. In wheel cylinder only the hydraulic pressure is converted in to mechanical work of brake shoes.

Construction

There are two movable pistons located inside the wheel cylinder. At the center position of these pistons inlet port is located. Nearer to this an bleeder screw is located. For avoiding oil leakages on the surroundings of the pistons, a rubber seal is fitted on it. For avoiding sticking of the pistons a return spring is fitted in between them. Moreover, to avoid entering of dust particles inside the cylinder a dust cover is fitted on both the sides.

Working principle

When the brake pedal is pressed by the driver, the brake fluid in the master cylinder is pressurized and enters in to the center of the wheel cylinders through the pipelines. This pressurized fluid pushes the pistons inside the cylinder in the opposite direction and hence the push rod which is connected to the pistons pushes brake shoes in the outward direction and allows the shoes to press the brake drum. By this way the brake is actuated and reduces the speed of wheels.

When the brake pedal is released, pressure in the wheel cylinder is reduced. Due to the action of the spring force both the pistons are moved towards the inward direction. Hence the brake fluid is returned back to the compression chamber of the master cylinder from the wheel cylinder through the same pipelines. The working principle of the hydraulic brake can be seen in Figure.

Bleeding system in hydraulic brake

In the hydraulic brake system if there is any air enters into any part of the braking system then the braking system will not functioning properly or totally. Thus the air must be removed. The process of removing the air present in the braking system is called as the bleeding system in hydraulic brakes. To release the air, the filler cap is opened

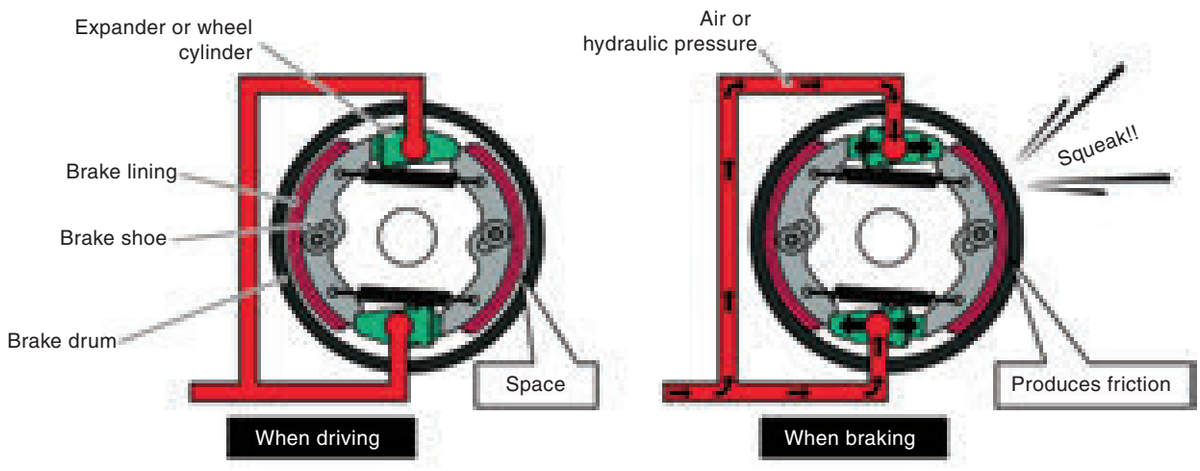


Figure 6.4.1.5 (b) Arrangement of the wheel cylinder

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